

Purnendu Tiwari

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Technical Skills

Languages: Java, JavaScript, TypeScript, Python, SQL, Bash

Web Development: React.js, Node.js, Express.js, REST APIs, Next.js, Prisma, MongoDB, MySQL, PostgreSQL

Cloud & DevOps: Amazon Web Services (AWS) – EC2, S3, CloudFront, IAM, VPC, CloudFormation, Docker, Docker Compose, GitHub Actions, CI/CD, Nginx, Cloudflare, Vercel, Linux, Linux Packaging (AppImage, Flatpak)

Developer Tools & Other: Git, GitHub, Bitbucket, Redis, BullMQ, Docker, PostgreSQL, Postman, WebRTC, Socket.IO

Experience

Software Development Intern

May 2026 - July 2026

FOSSEE, IIT Bombay

Mumbai, India (Remote)

- Collaborated with a mentor and a 3-member team to modernize the Linux packaging pipeline for the eSim application across **AppImage** and **Flatpak** formats
- Fixed recurring PyQt5 dependency crashes by bundling a self-contained Python runtime into the AppImage, eliminating the need for users to manually install matching host-side dependencies
- Resolved Qt platform plugin failures across **2 display server protocols (X11 and Wayland)** by building automatic detection logic, verified across **4 major Linux distributions (Ubuntu, Kubuntu, Fedora, Linux Mint)**
- Rebuilt the Flatpak offline install pipeline by fixing broken package sources and pre-validating dependency wheels ahead of build time, enabling fully offline, zero-network installs

Projects

sdcHouse – Real-Time Virtual Space Platform | Live | GitHub

- Developed a real-time multiplayer web application using **Next.js**, verifying state synchronization, AABB collision detection, and WebRTC handshakes with local sync latency under **10ms** across concurrent client sessions
- Engineered a **Socket.IO**-based signaling server on **Node.js/Express** to manage live game state (player positions, chat) and broker peer-to-peer connections between users
- Implemented **WebRTC** for direct peer-to-peer audio/video calls, removing media traffic from the server entirely by establishing connections directly via STUN

RCE Engine – Remote Code Execution Platform | GitHub

- Built a **Remote Code Execution (RCE) engine** that runs untrusted user code inside isolated **Docker** containers, enforcing a **128MB** memory cap, a **64-process limit** (fork-bomb protection), and no network access per submission
- Designed an asynchronous execution pipeline with **BullMQ** and **Redis** that decouples submission intake from sandboxed execution, threading a single UUID across the API, job queue, and database so any component can resolve a submission's state independently
- Created a **2-endpoint** REST API (**POST /submit**, **GET /submit/:id**) with **Express** and **Prisma/PostgreSQL** to accept code submissions and expose execution status and results
- Hardened the sandbox with defensive controls – execution timeouts with forced container kill, and a fresh throwaway container plus temp directory per submission – to safely execute arbitrary, untrusted code

Education

Vellore Institute of Technology (VIT)

B.Tech in Computer Science; CGPA: 8.99/10.0

Kendriya Vidyalaya No.1, Chhindwara

Higher Secondary (Class XII), CBSE – 92.4%

Bhopal, India

2023 - 2027

Chhindwara, India

2022

Certifications

AWS Certified Solutions Architect – Associate (SAA-C03) – Amazon Web Services

The Bits and Bytes of Computer Networking – Google (Coursera)

Achievements & Co-curricular

Top Coder: Cleared inter-campus qualifier, securing selection among the top 363 out of 10,000+ students

Leadership: President, Software Development Club; led a 6-member team to host 5+ tech events for 300+ students

LeetCode Knight: Achieved highest contest rating of 1852, ranking among the top 5% of coders on LeetCode

GirlScript Summer of Code: Ranked 419th among global open-source contributors

Music: Diploma in Classical Music (Raja Mansingh Tomar Music & Arts University)